HealthConnect Rwanda

Analytics Testing and Refinement Report (Week 7)

AnalystLab Africa Experience Lab, Data Analytics track | Yawa Jacqueline GBEDEVI | November 2026

1. From Week 6 to Week 7

Question	Answer
Main Week 6 output	A cleaned PostgreSQL database, 40 SQL queries, a six-page Power BI dashboard, an insights report and a slide deck.
Most important result	Billing and collection totals (9.31 billion billed, 90.1% collected) and the comparison of facilities on visits, revenue and satisfaction.
Main limitation	Only 5,000 of the 60,000 appointments referenced by billing exist, so facility, department and month views cover about 8% of bills.
What is tested	The KPI calculations, the values shown by the dashboard, and whether the Week 6 conclusions hold across segments.
Tracks involved	Data Analytics, with Data Science (feature validation) and Project Management (issue log).
Expected outcome	Confirm which figures are accurate, find conclusions that do not survive testing, and correct them.

The Week 7 brief mentions a clinic appointment dataset and attendance findings, which do not match the project of the previous weeks. Its testing tasks were applied to the Week 6 project, HealthConnect Rwanda Healthcare Analytics, whose objective is recommendations that improve service delivery, hospital operations, patient outcomes and financial sustainability.

2. Testing approach

Testing was done in three layers, each aimed at a different kind of error.

- **Calculation tests.** Thirty-two automated tests (script 04) recompute each KPI from the raw schema and compare it with the clean tables and the analytical views. Expected values never come from the objects being tested.
- **Dashboard tests.** Values shown in Power BI are compared with the SQL results. The dashboard was exported three times (HealthConnect_Analytics.pdf, HealthConnect_Analytics2.pdf and HealthConnect_Analytics_1.pdf) so that each correction could be re-tested.
- **Robustness tests.** Seven checks (B1 to B7 in script 04) plus a chi-square test on visit volumes ask whether the main findings hold across facilities, months, provinces and visit types, or could be chance. The p-values were computed from the query outputs.

3. Results

3.1 KPI and data validation

Of the 32 tests, 28 passed and 4 failed. The passes cover every row count (patients, appointments, billing, payments, claims, diagnoses, feedback, staff), the financial totals (9,309,401,961 billed, 8,389,454,556 collected, 919,947,405 outstanding, collection rate 90.12%), the appointment and billing figures for facility HF011, the average rating (2.987), the claim approval rate (49.98%) and all integrity checks (no duplicate keys, no unknown patients, doctors or departments, no impossible ages, no negative balances).

The four failures are informative.

- **T27, department label.** The count of departments labelled ICU is 0 instead of 1. The casing fix exists in script 01, but it was not applied to the delivered database, so the dashboard still reads Icu. The problem is cosmetic and remains open.
- **T30 and T31, billing status.** 4,042 bills marked Paid still carry a balance, and all 19,884 bills marked Pending have a payment recorded. The status label cannot be trusted.

- **T32, revenue summary.** The summary table totals 1.13 billion against 9.31 billion billed, an unexplained gap.

T30 to T32 are defects of the source data. They were expected to fail and are documented, not fixed.

3.2 Dashboard testing

The dashboard was exported three times and each export was compared with the SQL-validated figures. In the first run, 14 of the 20 initial checks passed. After the first round of corrections, five of the six failures were fixed, but re-testing exposed a side effect, a regression and a readability problem, so the second run stood at 18 of 23. After the second round, the final run passes 21 of 23 checks. Two items remain open, the province map and the readability of dark matrix cells (checks D21 to D23 were added during re-testing).

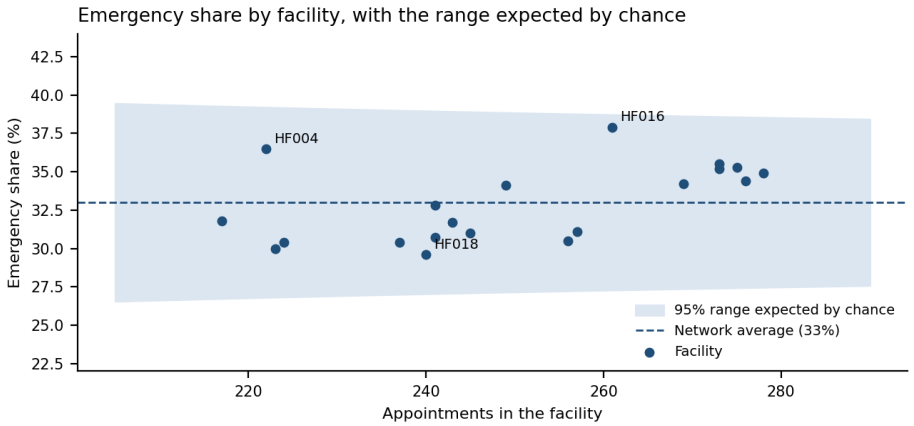
Two results stand out. Totals and facility-level values match the SQL exactly, and the blank facility row (8,538,268,522 billed, no appointments) equals the billing that has no appointment, which independently confirms the 8% coverage figure.

ID	Dashboard check	Expected (SQL)	Actual (final run)	Run 1	Run 2	Final	Notes
D1	Total patients (card)	20,000	20,000	PASS	PASS	PASS	
D2	Total appointments (card)	5,000	5,000	PASS	PASS	PASS	
D3	Total billed (card, matrix total)	9,309,401,961	9,309,401,961	PASS	PASS	PASS	
D4	Doctors, total facilities	350 and 20	350 and 20	PASS	PASS	PASS	
D5	Average rating card	2.99	2.99	PASS	PASS	PASS	Card showed 3 in run 1
D6	Gender split	50.45% and 49.55%	50.45% and 49.55%	PASS	PASS	PASS	
D7	Diagnoses in the tree	60,000, then 12,208 / 12,083 / 12,075 / 11,878 / 11,756	Same values	PASS	FAIL	PASS	Run 2: filter cut the tree to 5,000
D8	Staff by type	Nurse 700, Doctor 350, Admin 80, others 50, 40, 30	Bars match	PASS	PASS	PASS	
D9	Facility HF011	273, 44,206,632, 3.19, 61	Same values	PASS	PASS	PASS	
D10	Facility HF006	269, 43,624,779, 0.92, 64	Same values	PASS	PASS	PASS	
D11	Matrix row without facility	8,538,268,522 (bills without appointment)	8,538,268,522	PASS	PASS	PASS	Confirms 8% coverage
D12	Billed by department (Financial page)	Cardiology highest, Pharmacy lowest	Same order	PASS	PASS	PASS	
D13	Rating distribution	4,149 to 4,273 per score	4.1K to 4.3K	PASS	PASS	PASS	
D14	Bed utilization	0% (all beds Available)	0.00%	PASS	PASS	PASS	Known data limit
D15	Department matrix, billed (Cardiology)	86,248,476	86,248,476	FAIL	PASS	PASS	Run 1 showed 29,825,616
D16	Department matrix, appointments	5,000 across departments	5,000 (467 to 544 each)	FAIL	PASS	PASS	Run 1 showed 1,756
D17	Department matrix, staff	1,250 in total	125 per department, 1,250 total	FAIL	PASS	PASS	Run 1 showed 1,250 on every row
D18	Payment methods visual	Collected by method (Card 2.115 bn)	Collected, Card 2.12 bn	FAIL	PASS	PASS	Matches query 38
D19	Province map	Patients by province in Rwanda	Map opens on Africa, provinces located outside Rwanda	FAIL	FAIL	FAIL	Open. Province names ambiguous for the geocoder
D20	Decomposition tree, department level	Departments of linked diagnoses	Label No linked appointment (3,556), no blank	FAIL	PASS	PASS	Was 91% (Blank)
D21	Complaint trend by month (Poor and Excellent %)	Varies by month on linked reviews	Varies (about 33% to 48% poor)	PASS	FAIL	PASS	Run 2: date link to Feedback lost
D22	Readability of matrix cells	Legible values	Black text on dark cells (HF011 rating, Emergency)	n/a	FAIL	FAIL	Open. Contrast too low
D23	Rank on Total rows	Blank	Blank on both tables	FAIL	FAIL	PASS	Runs 1 and 2 showed 12 and 8

3.3 Robustness of the findings

Test	Result	Verdict
B1 Emergency share by facility	Range 29.6% to 37.9%. No z-score beyond 2 (largest 1.69). Chi-square p = 0.83.	Finding holds. Differences are chance.

Test	Result	Verdict
B2 Emergency share by month	Range 30.1% to 36.5%, all inside the chance band of about 4.5 points.	Finding holds. No trend.
B3 Satisfaction across facilities	ANOVA F = 0.735 (critical value about 1.59), p = 0.79.	Not supported. Facility differences are not significant.
B4 Revenue ranking, H1 vs H2	Rank correlation 0.24 (p = 0.30).	Not supported. The ranking is unstable.
B5 Top five facilities by half-year	H1 leaders HF011, HF013, HF006, HF001, HF002. H2 leaders HF010, HF006, HF003, HF014, HF013. Only HF006 and HF013 appear in both.	Not supported. The leaders change.
B6 Collection rate by segment	89.4% to 90.9% across provinces, 89.9% to 90.4% across visit types.	Finding holds. 90% is uniform.
B7 Query 14 grouping	All 350 doctors appear under more than one department.	Defect confirmed. Query corrected.
Extra: visit volume by facility	Chi-square 29.98 on 19 degrees of freedom, p = 0.052 (from query 12).	Borderline. Cannot be called a real difference.



Every facility sits inside the range expected by chance, which is why the 33% emergency share is described as a network-wide property and not as a facility problem.

4. Issues, refinements and re-tests

Component	Expected	Actual	First result	Issue and action	Re-test outcome
Appointment dates	0 rejected dates	864 of 5,000 rejected	FAIL	Future dates were treated as errors. Rule changed to accept the 2020 to 2030 range.	PASS. T24 and T28, 0 rejected
Outstanding amount	Billed minus collected	608.7 million (by status)	FAIL	Status-based logic ignored 311 million on Paid bills. Now computed from recorded payments.	PASS. T11, 919.9 million
Query 14, doctor workload	One row per doctor	Doctor split by department	FAIL	Appointment department is independent of the doctor (all 350 doctors affected). Query grouped by doctor only in script 03.	Corrected. Output not captured as evidence
Department label	ICU	Icu	FAIL	Text formatting changed the casing. Source casing kept in script 01, but not applied to the delivered database.	OPEN. T27 still fails, cosmetic
Department matrix (D15, D16)	Billed 86.2 M for Cardiology, 5,000 appointments	29.8 M, 1,756 appointments	FAIL	Appointment counts per department equalled the review counts, so the department filter reached appointments through the Feedback table. Relationships corrected.	PASS. 86,248,476 and 5,000 appointments
Staff by department (D17)	Staff split by department	1,250 on every row	FAIL	Department dimension was not related to staff. Relationship added.	PASS. 125 per department
Payment methods (D18)	Collected amount by method	Billed amount by method	FAIL	Visual changed to Total Collected.	PASS. Matches query 38

Component	Expected	Actual	First result	Issue and action	Re-test outcome
Decomposition tree (D20, D7)	Named departments, 60,000 at the root	91% (Blank)	FAIL	First fix, a filter on blank departments, applied to every level and cut the tree to 5,000 (D7 failed in run 2). Second fix, a label column renaming blank departments to No linked appointment.	PASS. Root 60,000, label shown
Complaint trend (D21)	Monthly variation	Flat 40% and 20% in run 2	PASS	Regression caused by the relationship fix, the date link to Feedback was lost. Relationship restored.	PASS. Varies by month
Rank on Total rows (D23)	Blank	Showed 12 and 8	FAIL	Rank measure changed to return blank outside facility and department rows.	PASS. Blank on Total rows
Colour rules and formats	Informative, readable	Green everywhere, red to green rating scale	FAIL	Thresholds and number formats corrected (percentage, 2.99), rating scale made neutral. Black text on dark cells is still hard to read (D22).	Partly resolved. D22 OPEN
Province map (D19)	Map of Rwanda	Blocked, then provinces placed outside Rwanda	FAIL	The visual was enabled, but names like Northern and Southern are geocoded elsewhere. A location column with the country, or latitude and longitude, would fix it.	OPEN. Recorded as a limitation
Facility revenue ranking	Stable ranking	Correlation 0.24 (B4)	FAIL	Presented as descriptive only, with the instability stated.	Wording updated in sections 5 and 6
Billing status, revenue summary	Consistent with payments and billing	Contradictory	FAIL	Source defects, documented and excluded from conclusions.	Unresolved at source
Emergency share, collection rate	Stable across segments	Stable (B1, B2, B6)	PASS	No issue found. Validation recorded.	n/a

5. Validated findings and updated insights

Findings that survived testing and can be stated with confidence. Totals and facility-level values in the dashboard also reproduce the SQL exactly.

- **Collections.** 9.31 billion billed, 8.39 billion collected (90.1%), 920 million outstanding (9.9%). The 90% rate is nearly identical across provinces and visit types, so the gap is systemic, not local.
- **Emergency load.** One visit in three is an emergency, in every facility and every month.
- **Insurers.** About half of all claims (50.0%) are approved and the rest are pending, with none rejected, at similar rates for the four insurers.

Findings that did not survive and are withdrawn or weakened.

- **Satisfaction by facility.** The ranking, and the idea that HF013 and HF014 are poorly rated, are not supported ($p = 0.79$).
- **Revenue ranking.** HF011 led in the first half of the year only. Rankings should not guide decisions until the full appointment history is loaded.
- **Staffing against demand.** The 28% gap in visits between facilities is borderline ($p = 0.052$), so a staffing imbalance is plausible but unproven.

6. Updated recommendations

Recommendation	Status after testing	Evidence
Tighten collections on the 11,887 bills with a balance	Confirmed	T09 to T12 pass, 90% rate stable across segments (B6).
Review the emergency pathway	Confirmed	33% share stable across facilities and months (B1, B2).
Set processing targets with insurers	Confirmed	Half of the claims are pending for all four insurers (T18).
Match staffing to demand	Conditional	Re-test on the full 60,000 appointments before moving staff (chi-square test, $p = 0.052$).
Investigate HF013 and HF014 on satisfaction	Withdrawn	Differences are not significant (B3).

Recommendation	Status after testing	Evidence
Fix data capture	Confirmed, higher priority	Missing appointments and contradictory status limit every facility-level conclusion.

7. Cross-track validation (HC-POD)

The Data Analytics findings were prepared for the Data Science and Project Management tracks. The part of the activity that belongs to this track is complete, because every fact handed over is backed by a test in this report. The part that depends on the other tracks had not produced a response at the time of writing.

Item	Record
Tracks	Data Science (modelling decisions) and Project Management (issue tracking).
Dependency	Validated findings and data limits that feed modelling choices and project decisions.
Output provided	Six validated facts for Data Science and eight issues for the Integration and Issue Log. Facts: 33% emergency share, 5,000 of 60,000 appointments, unusable phone column, insurance ID without an uninsured value, department and facility mismatch, unreliable billing status.
Testing done	Each fact is reproduced by a test, namely T29 (coverage), the phone distinct count, the 15 insurance codes, B1 and B2 (emergency share), the 95% department mismatch, and T30 and T31 (billing status). D11 independently confirms the 55,000 unlinked bills.
Finding	All facts hold. Together they show that facility-level results rest on about 8% of bills.
Action and change	The staffing recommendation was made conditional and the satisfaction-based recommendation was withdrawn, so no other track builds on conclusions that failed testing.
Output received	No response from the other tracks had been received at the time of writing, so no re-test of their side is recorded.
Evidence	clean.test_results, the robustness query outputs (B1 to B7) and the three dashboard exports.

8. Limitations and remaining issues

- **Resolved.** Date validation, outstanding calculation, doctor workload grouping, department matrix, staff by department, payment method measure, decomposition tree, complaint trend, rank on Total rows.
- **Open on the analytics side.** Province map (D19), readability of dark matrix cells (D22), ICU label (T27).
- **Unresolved at source.** Missing 55,000 appointments, billing status contradicting payments, revenue summary mismatch, corrupted phone numbers, department and facility mismatch.
- **Not possible with this dataset.** Admissions, laboratory, prescriptions, referrals, facility names and locations, bed utilization.
- **Testing limits.** Tests confirm that the pipeline reproduces the raw data faithfully. They cannot confirm that the raw data reflects reality. Satisfaction results rest on 1,756 reviews linked to appointments.
- **Before Week 8.** Apply the ICU casing fix to the database, fix the province map and the readability of dark matrix cells, capture the output of the corrected query 14, and record the responses of the other tracks.